

Atoms and Molecules:

- Around 500 B.C., Indian Philosopher named Maharishi Kanad postulated that matter can be divided into extremely small particles, which can't be divided further and are called as Parmanus.
- Greek Philosophers - Democritus and Leucippus gave same idea but called these indivisible particles as "Atoms".

Atom - "That can't be cut".

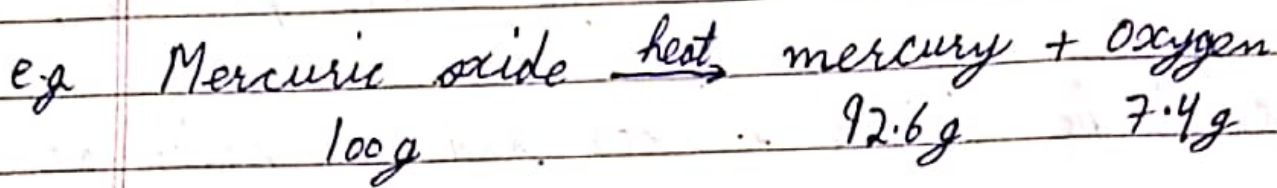
→ Laws of Chemical Combination:

- 1) Law of Conservation of mass: (By Antoine Lavoisier)
- During any physical or chemical change

the total mass of reactants is equal to total mass of products.

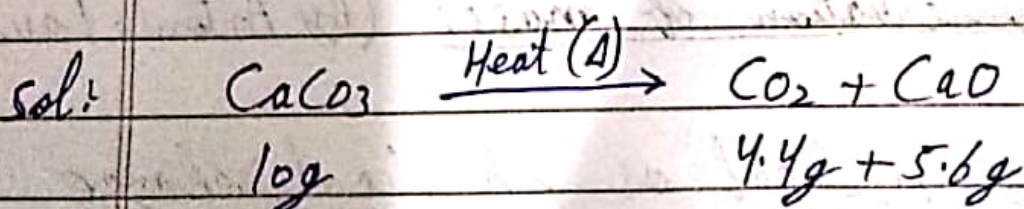
'OR'

"Mass can neither be created nor destroyed".



$$\begin{aligned} \text{Total mass of Reactants} &= \text{Total Mass of Products} \\ 100\text{g} &= 92.6 + 7.4 = 100\text{g} \end{aligned}$$

Q) If 10g of calcium carbonate (CaCO_3) on heating gave 4.4g of carbon dioxide (CO_2) and 5.6g of calcium oxide (CaO). Show these observations are in agreement with law of conservation of mass?



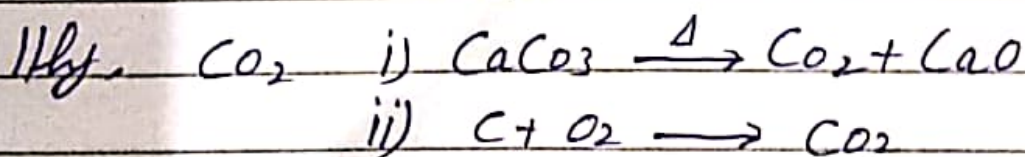
- Acc. to Law of conservation of mass of reactants = Total mass of products.

$$10g = 4.4 + 5.6$$

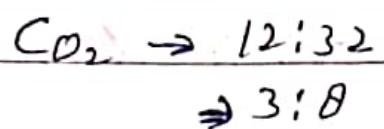
- Yes it follows Law of Conservation of mass.
- Law of Constant Composition 'OR' Definite Proportion
- ★ - Proposed by Joseph Proust.
- A pure chemical compound is always made up of same elements that are combined in same proportion by mass.

e.g. Water from any source - River, Lake, Ocean, Rain, etc. is always made up of Hydrogen and Oxygen in ratio H_2O

$$2:16 = 1:8$$



In case (i) and (ii) CO_2 prepared by different method but elements are and ratio is same.



- Q) When 1.375g of Cupric Oxide (CuO) was reduced on heating in a current of hydrogen the weight of copper (Cu) that remained was 1.098g. In another experiment 1.79g of copper was dissolved in nitric acid and resulting copper nitrate converted into cupric oxide by ignition. The weight of cupric oxide formed was 1.476g. Show that these results illustrate the law of constant proportion.

sol In experiment no. 1

$$\text{Mass of CuO} = 1.375 \text{ g}$$

$$\text{Mass of Cu} = 1.098 \text{ g}$$

$$\text{Mass of oxygen} = 1.375 \text{ g} - 1.098 \text{ g} = 0.277 \text{ g}$$

$$\begin{aligned} \% \text{ of Cu} &= \frac{\text{Mass of Cu}}{\text{Mass of CuO}} \times 100 = \frac{1.098 \times 100}{1.375} \\ &= 79.85\% \end{aligned}$$

$$= 80\%$$

$$\begin{aligned} \% \text{ of Oxygen} &= \frac{\text{Mass of oxygen}}{\text{Mass of CuO}} \times 100 = \frac{0.277 \times 100}{1.375} \\ &= 20\% \end{aligned}$$

In experiment no. 2

$$\text{Mass of CuO} = 1.476$$

$$\text{Mass of Cu} = 1.179$$

$$\text{Mass of oxygen} = 1.476 - 1.179 = 0.297$$

$$\begin{aligned} \% \text{ of Cu} &= \frac{\text{Mass of Cu}}{\text{Mass of CuO}} \times 100 = \frac{1.179 \times 100}{1.476} \\ &= 79.8\% \end{aligned}$$

$$= 80\%$$

$$= 80\%$$

$$\% \text{ of oxygen} = \frac{\text{Mass of oxygen}}{\text{Mass of CuO}} \times 100$$

$$= \frac{0.297}{1.476} \times 100$$

$$= 20\%$$

$\therefore$ Law of constant composition followed.

$\Rightarrow$ Dalton's Atomic Theory:-

- Proposed by John Dalton.

- John Dalton studied structure of matter and proposed following points:-

- 1) Matter is made up of extremely small particles called as Atoms.

- 2) Atoms of an element are similar in every respect, i.e; Mass, shape, size, etc.

- 3) Atoms of different elements are different in every respect i.e; mass, size, shape etc.
- 4) Atom is the smallest particle that takes part in a chemical reaction.
- 5) Atoms combine together in simple whole no. ratio to form molecules.
- 6) Atoms are indivisible i.e; they can't be divided further.

⇒ Limitations of Dalton's Atomic theory

- 1) This theory did not explain why the atoms of different elements are different i.e; this theory did not talk about internal structure of Atom.

2) Why and how atoms combine and form molecules

3) Atom is indivisible according to this theory but Atom can be divided.

⇒ Atom:- It is the smallest particle that may or may not have independent existence.

e.g. Atoms that exist independently = Na, C, K, B, etc.

Atoms that exist in combined form = O_2 , H_2 , N_2 , etc.

⇒ Symbols of Atoms of different elements:-

- John Dalton proposed some nortification to represent symbols of Atoms.

Atoms symbols.

e.g. Carbon → ●

Oxygen → O

Hydrogen → ⊙

- These symbols were later discarded.

⇒ Berzelius proposed that the symbols should include letters from name of atom.

- IUPAC - International Union of pure and applied chemistry; developed following symbols:-

A) Elements that use first letter of name as symbol:-

- i) Hydrogen - H
- ii) Oxygen - O
- iii) Nitrogen - N
- iv) Carbon - C, etc.

B) First two letters in symbols:-

Aluminium - Al

Helium - He

Lithium - Li

Silicon - Si

Neon - Ne

Argon - Ar, etc.

c) First and third letter of name:

Magnesium - Mg

Chlorine - Cl

zinc - Zn, etc.

d) Symbols from Latin Names:

Iron - ~~Ferr~~ Ferrum - Fe

Lead - Plumbum - Pb

Gold - Aurum - Au

Sodium - Natrium - Na

Potassium - kalium - K

Tin - stannum - Sn